# INVESTIGATIONAL PRODUCT & PHARMACY HANDLING MANUAL

## HEALEY ALS Platform Trial - Regimen G (DNL343)

Protocol ID: NCT05842941 | Version 2.0

This manual provides detailed instructions for the Investigational Drug Service (IDS) and clinical site staff regarding the receipt, storage, preparation, dispensing, and accountability of the Investigational Medicinal Product (IMP) for Regimen G of the HEALEY ALS Platform Trial.

## 1. Study Directory & General Information

| | |
|---|---|
| **Sponsor** | Denali Therapeutics Inc. |
| **Investigational Product (IMP)** | DNL343 (Active) / Matching Placebo |
| **Phase & Indication** | Phase 2/3 - Amyotrophic Lateral Sclerosis (ALS) |
| **CDMO Supply Vendor** | Fisher Clinical Services (Packaging, Labeling, Distribution) |
| **RTSM / IRT System** | Study-Specific Interactive Response Technology (IRT) Web Portal |

## Global Supply Chain Contacts

| Role / Department | Contact Information |
|---|---|
| **CDMO Clinical Supply Chain Manager**<br>For physical shipment, damaged kits, or TempTale issues. | Fisher Clinical Services Support<br>Email: ALS_RegimenG_Supply@fisherclinical.com<br>Phone: +1-800-555-1000 |
| **IRT / IWRS Helpdesk**<br>For system access, kit status changes, or unblinding emergencies. | Global IRT Support<br>Email: helpdesk@irt-provider.com<br>Phone: +1-800-555-2000 (24/7) |
| **Sponsor CMC / Pharmacy Liaison**<br>For compounding, stability, or tube feeding administration queries. | Denali Pharmacy Support<br>Email: pharmacy@denalitherapeutics.com |

## 2. Investigational Medicinal Product (IMP) Description

Regimen G utilizes a double-blind, placebo-controlled design for the initial 24-week period, followed by an open-label Active Treatment Extension (ATE) period where all subjects receive active DNL343.

| Parameter | DNL343 (Active) | Matching Placebo |
|---|---|---|
| **Formulation** | Granules for oral/enteral suspension | Granules for oral/enteral suspension |
| **Strength** | 100 mg per stick pack | 0 mg per stick pack |
| **Primary Packaging** | Aluminum foil stick pack (polyester/aluminum/low-density polyethylene laminate). | |
| **Secondary Packaging** | Cardboard carton containing 14 stick packs (segmented into 7 daily doses of 2 packets each). Labeling complies with Annex 13 and local regulations. | |
| **Dosing Regimen** | **200 mg Once Daily (QD).** Subjects consume two (2) stick packs simultaneously each day. | |

## 3. Shipping, Receipt, and IRT Registration

IMP is shipped via validated temperature-controlled couriers from the CDMO regional depots. Every shipment includes a **TempTale** continuous temperature monitor.

1. **Inspect Shipment:** Upon arrival, immediately inspect the outer shipping container for physical damage.
2. **Stop TempTale:** Press and hold the "STOP" button on the TempTale device for 3-5 seconds until the stop icon appears.
3. **Download Data:** Plug the TempTale USB into a computer. Save the automatically generated PDF report to your site's regulatory binder.
4. **Check for Alarms:** Verify if the TempTale triggered a temperature alarm.
    - *No Alarm:* Proceed to step 5.
    - *Alarm Triggered:* Immediately isolate the affected kits in the pharmacy under appropriate storage conditions. Mark them as "Quarantined" in the IRT and email the PDF report to the CDMO Supply Chain Manager for disposition. **Do NOT dispense.**
5. **IRT Registration:** Log into the IRT system. Enter the shipment number and confirm receipt of all physical kit numbers. The system will transition the kits from "In Transit" to "Available" inventory.

> **CRITICAL: Blinding Maintenance**
>
> During the 24-week placebo-controlled phase, the active and placebo stick packs are visually indistinguishable. Site staff must never attempt to unblind a kit. Dispense strictly by the Kit Number assigned by the IRT system.

## 4. Storage Requirements & Excursion Management

**Storage Conditions:** Both DNL343 and Placebo must be stored at **Controlled Room Temperature (below 30°C / 86°F)**. Protect from extreme heat and moisture. Kits must be stored in a secure, limited-access area (e.g., locked pharmacy) in accordance with ICH-GCP.

**Site Temperature Excursions:** If the pharmacy environmental monitoring system detects a temperature > 30°C:
1. Move the IMP to an alternate compliant storage location if the primary area cannot be rapidly corrected.
2. Quarantine the affected kits physically and systematically in the IRT.

## 5. Preparation, Compounding, and Dispensing

DNL343/Placebo is supplied as oral granules. Depending on the visit schedule (e.g., PK sampling days vs. take-home dosing), the IMP may be prepared in the clinic by site staff or prepared at home by the subject/caregiver.

> **In-Use Stability / Beyond Use Date (BUD):** Once the stick pack is opened and the granules are mixed with water or soft food, the dose **MUST be administered within 15 minutes**. Do not prepare doses in advance.

### Method A: Preparation in Water (Oral or Enteral Feeding Tube)

1. **Gather Supplies:** Two (2) assigned stick packs, a clean administration cup or dosing syringe, and room-temperature potable water.
2. **Empty Granules:** Carefully tear open both stick packs and empty the entire contents into the administration cup.
3. **Reconstitute:** Add approximately 30 mL to 50 mL of water to the cup. *[Standard CDMO Default – Sponsor to confirm exact minimum volume, especially for tube feeding restrictions].*
4. **Mix:** Swirl or stir gently until the granules are adequately suspended. (Note: Granules may not dissolve completely; a suspension is expected).
5. **Draw Up (Tube Feeding Only):** If administering via G-tube, J-tube, or NG-tube, draw the suspension into an appropriately sized enteral syringe.

### Method B: Preparation in Soft Food (Oral Administration)

1. **Gather Supplies:** Two (2) assigned stick packs and approximately 2 to 4 tablespoons of soft food (e.g., applesauce or yogurt). Do not mix with hot foods.
2. **Empty Granules:** Tear open both stick packs and empty the contents completely onto the soft food.
3. **Mix:** Stir the granules into the food until evenly distributed.
4. **Administer:** The subject must consume the entire portion of the mixture immediately (within 15 minutes). Fluid thickeners may be used concurrently if required for dysphagia.

### Dispensing to Subjects for Home Use

For take-home dosing, the Pharmacist or authorized Coordinator will dispense intact cartons to the subject.

- Log into the IRT to register the dispensing visit. The IRT will assign specific Kit Number(s) to the subject.
- Retrieve the exact matching Kit Number(s) from the pharmacy inventory.
- Affix a site-specific dispensing label to the secondary carton. The label must not obscure the original clinical label and must include:
  - Subject ID Number
  - Protocol Number (NCT05842941)
  - Date Dispensed & Dispenser Initials
  - Instructions: "Take 2 stick packs once daily. Store below 30°C."
- Provide the subject/caregiver with thorough training on the preparation methods (Water or Soft Food) and emphasize the 15-minute post-mixing administration window.

## 6. Administration Instructions (In-Clinic)

For visits requiring in-clinic dosing (e.g., Baseline, Week 4, Week 24 PK sampling):

- Ensure the subject is ready to receive the dose before opening the stick packs, due to the strict 15-minute stability window.
- **Post-Dose Flushing:** If administering via an enteral feeding tube, flush the administration cup/syringe with an additional 15–30 mL of water, draw it up, and flush the tube to ensure 100% of the dose is delivered and the tube remains patent. If administered orally via cup, add a small amount of water to the empty cup, swirl, and have the subject drink it to ensure no residual granules are left behind.

## 7. Accountability, Reconciliation, and Destruction

Under ICH-GCP, the Investigator is responsible for ensuring that all IMP is adequately accounted for throughout the trial.

- **Master Drug Accountability Log (DAL):** The Pharmacy must maintain a master log tracking the receipt, overall inventory, and final disposition of all cartons.
- **Subject Dispensing Log:** Maintain a subject-specific log tracking the exact dates and kit numbers dispensed to, and returned by, each subject.
- **Returns & Compliance:** Instruct subjects to return all unused stick packs, as well as all empty cartons and empty stick pack wrappers, to every clinic visit. Site staff must count the returned intact packets and wrappers to calculate dosing compliance and record this in the subject's eCRF and DAL.
- **Destruction:** IMP must NOT be destroyed on site without prior written authorization.
  - Used/Empty packaging may be destroyed on site according to standard hospital biohazardous waste policies *after* reconciliation is verified by the Clinical Research Associate (CRA).
  - Unused, expired, or quarantined kits must be returned to the CDMO depot (Fisher Clinical Services) using the pre-supplied return shipping materials, following IRT return authorization.

### 8. Action Items / Clarifications for Sponsor CMC Team

Prior to final sign-off and site distribution of this manual, the following operational gaps must be confirmed by the Sponsor Formulation/CMC team:

- **Reconstitution Volume:** Confirm the exact acceptable range of water volume (e.g., 30 mL - 50 mL) for reconstituting two 100 mg stick packs, specifically ensuring this volume is safe for subjects with fluid restrictions or specific enteral tubes.
- **Tube Compatibility:** Confirm if the DNL343 suspension is compatible with all standard enteral tube materials (Polyurethane, Silicone, PVC) and state the minimum recommended French (Fr) size to prevent tube clogging by the granules.
- **Dose Reductions:** The protocol permits a dose reduction to 100 mg QD (1 stick pack). Confirm if the IRT handles this by simply instructing the subject to consume 1 packet from their existing 14-packet carton, effectively doubling the duration of the kit, and how this impacts IRT resupply algorithms.